

Build the Report

Spotify Top-50-World — Assembling the 5 Power BI Pages

This document picks up exactly where the measures work ends. It assumes your data model and all 46 measures already exist — this is purely about where each one physically goes on the page.

1. Before You Start

Do not begin this document until every item below is true. Placing visuals on top of a model that isn't finished just means redoing the layout later.

- The star schema is fully built: `dim_songs`, `dim_dates`, `fact_chart_entries`, all uploaded and live in Supabase.
- Both relationships exist in the Power BI model (`dim_songs` ↔ `fact_chart_entries` on `song_key`, `dim_dates` ↔ `fact_chart_entries` on `date`) and are confirmed as clean 1-to-many, not many-to-many.
- `dim_dates` is marked as the Date Table.
- All 46 measures exist in the model — check the Fields pane, not just that DAX query view ran without errors.
- You've spot-checked at least a handful of measures on a blank card visual and gotten sensible numbers back, not blanks or errors.

If any box above is unchecked

Stop here and go finish that first. Every page in this document assumes a working, queryable model — without it, you'll be dragging fields that don't exist yet into visuals that will just show blank.

2. Your Backgrounds

Two background designs, used across all 6 report pages (1 cover page + 5 content pages):

Background	Description
BG-1 — Cover / Landing	The dark title-screen background: logo and wordmark on the left, album-art collage on the right, and the 4-button nav row (Home, Overview, Artists, Songs) beneath the logo. Used once, as the report's opening page. Nothing else gets placed on this page — it's a title screen, not a content page.

Background	Description
BG-2 — Grid Dashboard	The tile-based layout. This single design is reused as the background for every content page — same tiles, same header (logo + nav row), just different visuals dropped into the tiles and a different nav button highlighted as active on each page's copy.

BG-2's Named Slots

Every content page below refers back to these slot names — learn this table once, and every page section reads the same way.

Slot	Where It Sits	Shape
Header	Top strip, full width	Fixed — logo + nav, never edit
Slot 1	Far left, full height	Tall vertical panel
Slot 2A-2D	Second column, top	Four small stacked tiles
Slot 3	Second column, below the stack	Wide tile
Slot 4	Third column, top — the green tile	Large square-ish “hero” tile
Slot 5	Third column, below the hero	Wide tile
Slot 6A-6D	Right side, top	Four small/medium tiles, 2x2
Slot 7	Right side, middle	Large wide tile
Slot 8	Right side, bottom	Large wide tile

Not every slot gets used on every page

14 slots is more than most pages need. Where a page's table below marks a slot as “(unused)”, delete that tile shape from that page's own copy of the background rather than leaving an empty box — an unused tile reads as a mistake, not a deliberate choice, unless it's actually removed.

3. Making the Nav Buttons Actually Work

The Home / Overview / Artists / Songs buttons are part of the background image — they don't do anything on their own. To make them real navigation, on every page:

1. Insert a blank button (Insert > Buttons > Blank) directly on top of each nav pill in the background image, resized to match it exactly.
2. Set its fill and border to fully transparent, so the background's own button graphic shows through underneath.

3. In the Format pane, turn on Action, set Type to Page navigation, and set Destination to the target page.
4. Repeat for all 4 pills, on every page that has this header — yes, this means placing the same 4 invisible buttons on each content page.

On the 2 pages without a matching nav button

Content & Release-Format Trends and Seasonality & Chart Behavior don't have a corresponding pill in this nav design. Still place the same 4 invisible buttons on these two pages for visual consistency (so the header behaves identically everywhere), but reach these two pages through Power BI's own page tab strip at the bottom of the window, not through this custom nav bar.

4. Page-by-Page Assembly

Six pages total. Each section below names the background, the active nav state, and exactly which visual goes in which slot.

Page 0 — Cover

Background: BG-1

No visuals, no measures. This page is the background image itself, plus the 4 invisible navigation buttons from Section 3. If you want a report title or your name/date somewhere, a single text box in the open dark space above the nav row works — keep it small and out of the way of the logo and album art.

Page 1 — Executive Overview

Background: BG-2 · Nav: Overview

Slot	Visual	Type	Measure(s) / Field(s)
Slot 4	Total Songs	Big number card	[Total Songs]
Slot 1	Pct Explicit Songs	Card / gauge	[Pct Explicit Songs]
Slot 2A	Distinct Songs	Card	[Distinct Songs]
Slot 2B	Distinct Artists	Card	[Distinct Artists]
Slot 2C	Distinct Song-Artist Combos	Card	[Distinct Song-Artist Combos]
Slot 2D	Position 1 Artists	Card	[Position 1 Artists]
Slot 3	Popularity Range	Multi-field card	[Max Popularity] + [Min Popularity]
Slot 5	Avg Position	Card	[Avg Position]
Slot 6A	Avg Popularity	Card	[Avg Popularity]
Slot 6B	Position 1 Songs	Card	[Position 1 Songs]
Slot 6C	Explicit Songs	Card	[Explicit Songs]
Slot 6D	(unused — delete)	-	-
Slot 7	(unused — delete)	-	-
Slot 8	(unused — delete)	-	-

Every visual on this page is a Card — no charts, no tables. It's meant to be readable in five seconds, not studied.

Page 2 — Artist Deep Dive

Background: BG-2 · Nav:
Artists

Slot	Visual	Type	Measure(s) / Field(s)
Slot 1	Artist Table	Table	Rows: dim_songs[artist]. Columns: [Songs per Artist], [Distinct Songs per Artist], [Avg Popularity per Artist], [Position1 Hits per Artist], [Artist Rank by Chart Entries], [Pct of All Entries by Artist]
Slot 4	Artist Catalog Dominance	Treemap / Donut	Group: dim_songs[artist]. Values: [Distinct Songs per Artist]. Top N filter: 10
Slot 3	Top 10 Artist Entries	Horizontal bar chart	Axis: dim_songs[artist]. Values: [Top 10 Artist Entries]
2A-2D, 5, 6A-6D, 7, 8	(unused — delete all)	-	-

Only 3 visuals on this page, but two of them (the table and the treemap) are dense — that's exactly why they get the two largest slots (the tall panel and the hero tile) instead of being squeezed into small ones.

Page 3 — Song Lifecycle & Trends

Background: BG-2 · Nav:
Songs

Slot	Visual	Type	Measure(s) / Field(s)
Slot 4	Chart Longevity Leaders	Horizontal bar chart	Axis: dim_songs[song]. Values: [Days Charted]. Top N filter: 10-15, sorted descending
Slot 1	Song Detail Table	Table	Rows: dim_songs[song], dim_songs[artist]. Columns: [Peak Position], [First Chart Date], [Last Chart Date], [Days Charted], [Avg Days Since Release at Charting], [Position Range], [Position StdDev]
Slot 5	Fastest Climbs to #1	Scatter chart	X: [Days to No1]. Y: [Days Charted]. Size: [Avg Popularity]. Legend: dim_songs[song_key]
Slot 2A	Avg Duration Minutes	Card	[Avg Duration Minutes]
Slot 2B	Max Duration Minutes	Card	[Max Duration Minutes]

Slot	Visual	Type	Measure(s) / Field(s)
Slot 2C	Min Duration Minutes	Card	[Min Duration Minutes]
Slot 2D	<i>(unused — delete)</i>	-	-
3, 6A-6D, 7, 8	<i>(unused — delete all)</i>	-	-

A blank scatter is not a broken scatter

Songs that never hit #1 will show no bubble on the Slot 5 scatter chart, since [Days to No1] deliberately returns BLANK() for them. Don't chase this as a bug — it's correct behavior, already covered in the measures documentation.

Page 4 — Content & Release-Format Trends

Background: BG-2 · Nav: none

Slot	Visual	Type	Measure(s) / Field(s)
Slot 4	Duration vs. Popularity	Scatter chart	X: [Avg Duration Minutes]. Y: [Avg Popularity]. Legend: dim_songs[is_explicit]. Trendline: on
Slot 5	Singles vs. Albums Over Time	100% stacked column	Axis: dim_dates[year_month]. Values: [Singles Count], [Albums Count]
Slot 2A	Explicit Songs	Card	[Explicit Songs]
Slot 2B	Non-Explicit Songs	Card	[Non-Explicit Songs]
Slot 2C	Avg Popularity Explicit	Card	[Avg Popularity Explicit]
Slot 2D	Avg Popularity NonExplicit	Card	[Avg Popularity NonExplicit]
Slot 6A	Avg Tracks per Album	Card	[Avg Tracks per Album]
Slot 6B	Album Type Count	Card	[Album Type Count]
Slot 6C	Pct Singles	Card	[Pct Singles]
Slot 6D	Popularity Std Dev	Card	[Popularity Std Dev]
1, 3, 7, 8	(unused — delete all)	-	-

This page fills both stacks (2A-2D and 6A-6D) as an 8-card comparison block flanking the two charts — the most card-dense page in the report, which is appropriate given how much of Section 4's EDA findings live here.

Page 5 — Seasonality & Chart Behavior

Background: BG-2 · Nav: none

Slot	Visual	Type	Measure(s) / Field(s)
Slot 4	Yearly Trend	Multi-line chart	Axis: dim_dates[year]. Values: [Songs per Year], [Avg Popularity per Year]
Slot 5	Explicit Content Over Time	Line chart	Axis: dim_dates[year_month]. Values: [Pct Explicit per Month]

Slot	Visual	Type	Measure(s) / Field(s)
Slot 3	Day-of-Week Chart Behavior	Column chart	Axis: dim_dates[weekday_name] (sort by weekday_num, not alphabetically). Values: [New Entries Count]
Slot 2A	Weekend Entries Count	Card	[Weekend Entries Count]
Slot 2B	Avg Duration per Year	Card	[Avg Duration per Year]
Slot 2C, 2D	(unused — delete)	-	-
1, 6A-6D, 7, 8	(unused — delete all)	-	-

The lightest page of the five, both in measure count and in slots used — that's a fair reflection of the underlying analysis, not a sign something's missing.

5. Final Assembly Checklist

Before calling the report done, confirm every item below:

- All 6 pages exist: Cover, Executive Overview, Artist Deep Dive, Song Lifecycle & Trends, Content & Release-Format Trends, Seasonality & Chart Behavior.
- Every unused tile on every content page has been deleted, not left empty.
- All 4 invisible nav buttons are placed and working on every page that uses BG-2, including the 2 pages without a matching pill.
- Every visual shows real numbers — no blank cards, no “value cannot be determined” errors.
- The 2 pages without a nav button are reachable via Power BI's own page tab strip.

Once every box is checked, the report is functionally complete. What's left after this is polish — consistent number formatting, tooltips, and visual titles — not structure.